

# Model Performance Comparison

Team 10

章明阳 2022201378

| Metrics    | In sample  | Out of sample | Cross-validation | Kaggle Score |
|------------|------------|---------------|------------------|--------------|
| OLS        | 890,947.53 | 868,977.13    | 892,021.94       | 25.91        |
| LASSO      | 930,039.25 | 902,617.61    | 1,163,859.95     | 26.36        |
| Ridge      | 890,947.38 | 868,976.94    | 892,003.55       | 29.62        |
| ElasticNet | 899,823.14 | 875,978.78    | 1,159,477.36     | 32.33        |

林琪岷 2023200211

| Metrics    | In sample   | Out of sample | Cross-validation | Kaggle Score |
|------------|-------------|---------------|------------------|--------------|
| OLS        | 100767.7687 | 127051.5078   | 127051.5078      | 53.91        |
| LASSO      | 100058.5344 | 125663.5788   | 121144.2602      | 53.91        |
| Ridge      | 100159.7700 | 126145.8116   | 121646.3263      | 53.91        |
| ElasticNet | 102605.3111 | 124567.4325   | 122795.4957      | 53.91        |

刘永泽 2023201776

| Metrics | In sample | Out of sample | Cross-validation | Kaggle Score |
|---------|-----------|---------------|------------------|--------------|
| OLS     | 323,500   | 323,400       | 446,500          | 9.35         |
| LASSO   | 323,500   | 323,400       | 446,400          | 8.87         |
| Ridge   | 323,500   | 323,400       | 446,500          | 12.89        |
| Elastic | 323,500   | 323,400       | 446,500          | 14.63        |

## Final Combined Performance

| 成员  | 模型         | In sample  | Out of sample | Cross-validation | Kaggle Score |
|-----|------------|------------|---------------|------------------|--------------|
| 章明阳 | OLS        | 890,947.53 | 868,977.13    | 892,021.94       | 25.91        |
| 章明阳 | LASSO      | 930,039.25 | 902,617.61    | 1,163,859.95     | 26.36        |
| 章明阳 | Ridge      | 890,947.38 | 868,976.94    | 892,003.55       | 29.62        |
| 章明阳 | ElasticNet | 899,823.14 | 875,978.78    | 1,159,477.36     | 32.33        |
| 林琪岷 | OLS        | 100,767.77 | 127,051.51    | 127,051.51       | 53.91        |
| 林琪岷 | LASSO      | 100,058.53 | 125,663.58    | 121,144.26       | 53.91        |
| 林琪岷 | Ridge      | 100,159.77 | 126,145.81    | 121,646.33       | 53.91        |
| 林琪岷 | ElasticNet | 102,605.31 | 124,567.43    | 122,795.50       | 53.91        |
| 刘永泽 | OLS        | 323,500    | 323,400       | 446,500          | 9.35         |
| 刘永泽 | LASSO      | 323,500    | 323,400       | 446,400          | 8.87         |
| 刘永泽 | Ridge      | 323,500    | 323,400       | 446,500          | 12.89        |
| 刘永泽 | Elastic    | 323,500    | 323,400       | 446,500          | 14.63        |